DHL Transport Label
Automated Label Specifications

Automated label specifications
version 2.5.2

This document is normative and relevant for all units of DHL Express & DHL Freight globally :

DHL Danzas Air & Ocean is in the process of being integrated:

DHL
De Kleetlaan 1
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Belgium

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Change Control

This document complies with change control procedures described in the L&I Change Control Guideline 1.0. In general, only significant changes to the solution and concept detailed in this document will trigger a new release version.

Version	Date	Changes
1.0	2001-01-31	• Initial creation
1.1	2001-03-07	• CMR logo not mandatory • GTB reference removed • Customs clearance flag added to routing barcode • Only License Plates allowed that produce barcodes shorter than 90mm • Only sans-serif fonts allowed • IFTMIN message structure in 2D barcode updated • 2D barcode message examples updated
1.2	2001-03-20	• CMR logo removed • Sorting relevant product features in reverse video, if supported by printer • Shipment date added • Incoterms added to product features section • Maximum of 0.51mm added for the linear barcode narrow element dimension • Routing barcode encoded using EAN data-identifiers
1.3	2001-05-16	• “Parcel” changed to “Piece” • Incoterms removed from product features section

1.4	2001-07-20	• Document history added • CEN data-identifiers used for all barcodes • All barcodes limited to 91 mm length • Routing barcode modified • No blanks allowed in the barcode data, blank is used as CEN field separator • Product names extended • “Piece” used instead of “parcel”, “package” etc • CCD License Plate recognition bar added • Reference to UPU standard S26-4 added • Definition of customs clearance goods added • Font sizes changed • Layout revised • Routing barcode not mandatory for domestic when 2D barcode is used • Barcode restrictions and plausibility checks added • Customer logo added • “From” and “To” in English/Bilingual for international shipments • Label can be split over two smaller labels • CMR logo optional • In the future also an EAN data-identifier will be allowed for the routing barcode • Code 39 also allowed • Label with routing information only added

1.5	2001-08-07	• Separating lines added in lay-out • Label examples updated with new routing barcode content • IAC in LP in label examples updated • Product added to the routing label • Examples of maximum length License Plates added • Position of shipment date changed • Code for C.O.D. and EXW changed • Partner specific depot-code added to product features section • Example of maximum length barcodes with quiet zones explained • Issuing agency code changed to “JD00” • IAC added to plausibility check on License Plate • Label examples updates with IAC “JD00”

1.6	2001-09-13	• Partner specific routing chapter left empty • 2D barcode content chapter left empty • Plausibility checks on LP extended • Font size of ship to address country and postcode changed • Shipment block changed (customer no., shipment date optional, text “pieces” removed) • In the from- and to addresses the ISO-3166 country code instead of the country name is also allowed • Check on multiple 2D barcodes on the piece changed • Partner specific routing section in product feature line removed • DPEE standard routing information changed • Shipment information block added to the routing label • Postcode in routing barcode max A9 • Product code in routing barcode equals Alice code with leading zero • Deviating requirements for CAS systems are marked (whole document; see symbol below • Number of lines for consignee address has been specified more precisely • The font size table has been added by equivalent values for “height in mm” and “size in dots at 200 dpi” • Separator to be used in case of more than one sorting relevant product features has been defined • Product feature coding explanation added by examples
2.0	2001-10-05	• Code 39 explicitly excluded for CAS and DPEE internal systems • Minimum size of inverse box for handling relevant features • Correction: term “comma” instead of “colon” for feature separator • DP EE partner specific routing info added as optional section on routing label • Number of lines in address fields specified more precisely • Usage of ISO country code for sender address confirmed/clarified • Data identifier not part of License Plate length • DPEE Partner logo in words if printer does not support images for CAS • Terms and definitions updated

2.1	2001-11-28	• Examples for the routing barcode added • Error correction: DI in routing code is A2 instead of A3 • CMR logo is mandatory • Shipment date not mandatory in examples • Product codes for EUROPACK and EUROPREMIUM changed • Customer number in example of all sections • License Plate valid characters clarified (Returns to alternative address clarified • Partner specific routing barcode digits only domestically or bilaterally agreed • Optional version no. added Optional GTB (general terms of business) reference added • Chapter 11 and 12 swapped • Error detection in barcodes explained • Check digit in human readable representation of License Plate allowed according to UPU standard S26 • License Plate examples added • Partner specific routing information and article information in separate blocks for clarity (Separation of License Plate from DI and AI clarified ANS MH 10 replaced by ANSI • CEN License Plate replaced by ISO License Plate
2.2	2001-12-11	• Label length definition clarified Table of product features not shown on label replaced by reference to master data • Recommendation added to use barcode height of at least 28 millimeters. • Field separator FNC1 introduced to separate partner specific part of routing code • Optional DP EE Partner specific routing codes max length N6 • Error corrected: “delivery time coding” instead of “delivery day coding” • Modulo 10 check digit scheme included • Order of country code and postcode permuted in routing barcode • Index added ()

2.3	2002-12-18	• Address fields: Country name usage corrected • Reference to IFTMIN Message Implementation Guide for PDF417 content has been added • Explanations for normative references have been extended • Barcode length calculations corrected and further explanation added • Text concerning EAN and ANSI organisation deleted • Table of product codes: Reference to a separate file added • Data Identifier 5J and 6J added • Ship to address : The sentence “please refer to separate address label on piece” • Chapter “Landscape orientation” deleted. • The sorting information has a header “Handling” deleted • International Address structure added. (Handling information in a surrounding frame with the header “Handling” information printed left aligned • Plausibility checks for the routing barcode updated (EAN Routing content examples added • New routing label • Label example with EAN 128 routing and License Plate barcode added • Encoding EAN 128 routing barcode added
2.4	2003-01-29	• The document now covers the DHL Transport Label variants of Deutsche Post Euro Express and Danzas Eurocargo; wording adapted
2.41	2003-02-03	• New DHL logos inserted as images • "Survey of measures and sizes" inserted • Example label with manual sorting section inserted

2.5	2003-08-22	• Reference to standards added to concept EAN SSCC structure adapted • License Plate length examples: corrected, EAN example added • Barcode symbologies: Application Identifier added • Company name corrected: EAN International • Reference to General UCC/EAN specifications and ISO 15438 (PDF417) added • Definition of EAN 128 adapted • PDF417 usage description re-arranged • reference to ISO 15434 and ISO 15438 added • Mandatory structure of the Routing label added • Reference to the to the common standard DHL Express product code file added • Origin Service Area Code for air-based shipments added • Air-based shipment information added • Warsaw Convention / DHL Terms & Conditions for air-based shipments added • Conversion table of x-dimension measure units added • AWB Code 39 Bar Code Specification added • Air –based Network label added (NSS 21.3.1) • Mandatory for air-based shipments to print out of License Plate in groups of four digits • Checksum for the LP information in text form. • IAC table updated • Label with customer specific barcode and article information added. • Label certification added • Capture) Label examples for Air-based shipments added. • Special A5 air format added • Archival Label copy for air products added

2.5.01	2004-05-10	• Local Temporary Exceptions added to align Air Express Network Service Standards **(10)** • Overall Document Layout & Structure of the document has been modified • Section on Target Groups added **(1.1)** • Chapter on change reasons added **(1.3)** • Terms “section” and “segment” added **(4.1)** • Mandatory, Conditional, Optional label sections identified **(4.1)** • Routing information chapter rewritten **(5)** • License Plate chapter redrafted **(6)** • Chapter on Master Data added **(1.2)**

2.5.2	2004-06-25	• DHL Logo to be put on the top right side of all labels as per Corporate Design specs. • DHL Express replaced by DHL • Copyright address changed from Bonn to Brussels (see cover page) • Cover page text slightly amended (see cover page) • The term Common Label replaced by DHL Transport Label • The term Item replaced by Piece **(3.4)** • The term zipcode replaced by postcode **(4.4.8)** • Warsaw Convention legal text has been moved to the legal section together with the GTB legal text. **(4.4.5)** • Definition of Date feature added **(5.7.2)**. • Comma as a feature separator has been removed **(5.7.4)** • Glossary updated **(18)** • IAC list updated **(16)** • Normative references restructured **(17)** • Index updated **(19)** • Handling features: the comma separator has been removed, only the minus sign should be used**. (5.7.4)** • US legal text added **(4.4.5)** • Label examples updated **(9)** • Routing Label Layout updated **(5.10.3)** • Routing Label : LP not mandatory anymore **(5.10.3.1)**

Reviewed by

Department	Date	Status
Engineering & Infrastructure Global		open
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ACI Global		open
GIS		open
Information Infrastructure Global		open
Customer Operations		open

Approved by

Boards	Date	Status
GMEG		open
EUROMEG		open
ROMG		open

Symbols used in this document

Parts that are valid for CAS system only are marked with a CAS symbol.

Cross references are marked with this symbol. They are provided to indicate related chapters or to refer to associated information..

Important remarks are marked with a red/dark exclamation mark.

Product-specific information is marked with this symbol

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1 Scope

This document covers the DHL specifications of the automated (non-preprinted) version of the DHL Transport Label.

This transport label is to be used by all units in the DHL Worlwide Express network, in all domestic and international operations, as far as the respective Business Unit has been considered by this document already.

The DHL automated transport label is defined according to the world-wide standard **ISO 15394** and its European predecessor **CEN EN1573** *Multi Industry Transport Label*.

Regional specifications extending this document or putting parts of it in concrete terms are possible but must by no means contradict this document.

For the lay-out of pre-printed labels/waybills please see the **Manual Waybill Specifications.**

1.1 Target Groups

The present internal version 2.5.2 is more appropriate for DHL internal purposes, i.e. mainly operations and customer automation.

The document has been divided into two main sections.

The first section called "**Core DHL Transport Label and Identifier Specifications"** details the DHL specific label layout requirements whereas the second section called "**Background information**" provides more external and/or technical information.

As a consequence, the document can also be used without the section2.

A shorter external version is being developed specifically for use by external customers.

1.2 Master Data

The Master Data Document is a central data repository which functions as an essential appendix of the present document. While the current document displays the baselined concept and specifications of the DHLTransport Label, the Master Data contains a regular update of the description of the elements which appear on the label and which are subject to regular changes.

It is therefore imperative to note that some of the product and feature names and codes used in the present document may not be relevant anymore at the time the document is consulted.

It should also be highlighted that not label related referenced data displayed on the label are centrally owned and maintained by the Label Department.

1.3 Change Reasons

The version 2.5.2 does not aim at changing the main concept or solution described in the 2.5 version, the main change reasons are twofold :

- The structure of the 2.5 version needed to be changed in order to display a more consistent and logical organisation of the document sections whereby an attempt has been made at differentiating between DHL label & identifier specifications which are detailed in the first core section and external standards and supporting information which have been moved to a dedicated appendix section.
- Based on feedback from countries, other regions and related applications, agreements have been reached to include regional temporary exceptions to the 2.5 solution. These exceptions which have also been integrated in the NSS version 21.3.1 are expected to be absorbed in the 2.6 version.

1.4 Other reference documents

Below listed documents can be used as a valuable reference for anyone who wishes to obtain additional internal or external information.

- **NSS 21.3.2.0 (Former Air Express Network Service Standards : Machine Printed Shipment Labels)**
- **DHL Glossary** (see chapter 18)
- **International Standards** (see chapter 17)

2 SECTION A : CORE DHL TRANSPORT LABEL AND IDENTIFIER SPECIFICATIONS

3 Clarification of concepts

3.1 Principles

The purpose of the DHL DHL Transport Label is to facilitate the automatic exchange of data among all members within the channel of distribution: Sender, DHL units and receiver.

3.2 DHL Transport Label

The DHL transport label refers to an adhesive piece of paper attached to a transport unit which functions as a carrier of information relevant to the handling and delivery of the transport unit.

The DHL Transport label is used for transportation and issued in accordance with global DHL label specifications and ISO 15394 and 15459.

However, the DHL Transport Label does not replace the need for official **transport order data documents**.

3.3 Shipment

A Shipment (or consignment) is an administrative concept that is used to designate

- one or more pieces that are
- to be transported at the **same** time, with the
- **same** product and features, from the
- **same** location to the
- **same** delivery address

The shipment is identified by a unique shipment number, assigned by the responsible DHL unit.

The referencing of the pieces to the shipment has to be given to the transport provider in form of a document (waybill or manifest) or in electronic form.

3.4 Transport unit (piece)

Within the networks of DHL, a transport unit is the smallest identifiable unit suitable for shipping, i.e. a pallet, carton, bag etc. A transport unit will be referred to as a piece in this document.

3.5 License Plate

According to ISO 15394, the License Plate barcode should be placed as the last segment at the bottom of the label.

The License Plate is a unique transport unit identifier assigned to each individual piece according to ISO15459. It is not allowed in any situation to have one piece labeled with two different License Plate numbers.

A License Plate can be produced from any source and used in transportation with DHL

The License Plate is the key providing access to information stored in computer files and which may be transmitted by EDI. It enables the tracking and tracing of individual pieces.

It must be guaranteed that only one such identifier exists in the whole world.

3.6 One-dimensional barcode

Traditional barcode technology is sometimes referred to as one-dimensional, or a linear barcode. In this document, these concepts are synonymous.

3.7 Two-dimensional barcode (2D-code)

This barcode technique involves reading the symbol both vertically and horizontally in order to interpret all the data content. PDF417 is the main two-dimenional barcode referred to in this document.

3.8 Goods

Generic term used to describe all physical things accepted by DHL for transportation.

3.9 Liability

It is assumed that the DHL Transport Label is usually printed and applied to the transport piece by the customer/sender who is responsible for the correctness of all the information on the label.

3.10 International Standards

To facilitate the implementation of the DHL Transport Label by both customers and Supplier of label printing applications, most parts are based on ISO standards or on common industry standards.

See chapter on Normative References for a detailed list of standards applied to the DHL Transport label solution.

Please also see the chapter on normative references **(17)**

4 Label Layout

4.1 Basic Structure

The numbered sections of the DHL Transport Label as shown below refer to the following paragraphs where the sections are described in more detail. The dimensions of the sections are described in a dedicated chapter.

Sections may be separated by horizontal lines to increase readability, however this is not mandatory.

The label consists of **sections** which may further contain one or more **segments**

As a general principle, any insecure or unreliable label content information should not be printed on the label. It is better to print nothing than to print wrong information.

This chapter defines which content of the sections and segments are **mandatory, conditional** or **optional.** ,

- **Mandatory** : means it must always appear independent of the situation. If a mandatory information is not available, the whole label or a particular label section/segment should not be printed. (E.g.: no DHL Transport Label should be printed without License Plate barcode)

  (Mandatory sections/segments are highlighted in **red**)

- **Conditional** : means it may be mandatory in somes cases and optional in other cases. (E.g. the CMR logo is only mandatory for international road-based transportation)

  (Conditional sections/segments are highlighted in **black**)

- **Optional** : means that the label field content is not mandatory, it does not have to be displayed on the label. (E.g.: the customer logo)

  (Optional sections/segments are highlighted in **blue**)

Note : For the routing information, the optional content information does not always indicate that the dedicated section or segment field remains empty/blank when printing the label.

For instance, if the optional delivery date or time features are not chosen, optional implies that zeroes are displayed in the routing barcode.

Please also see the chapter on routing barcode **(5.2)**

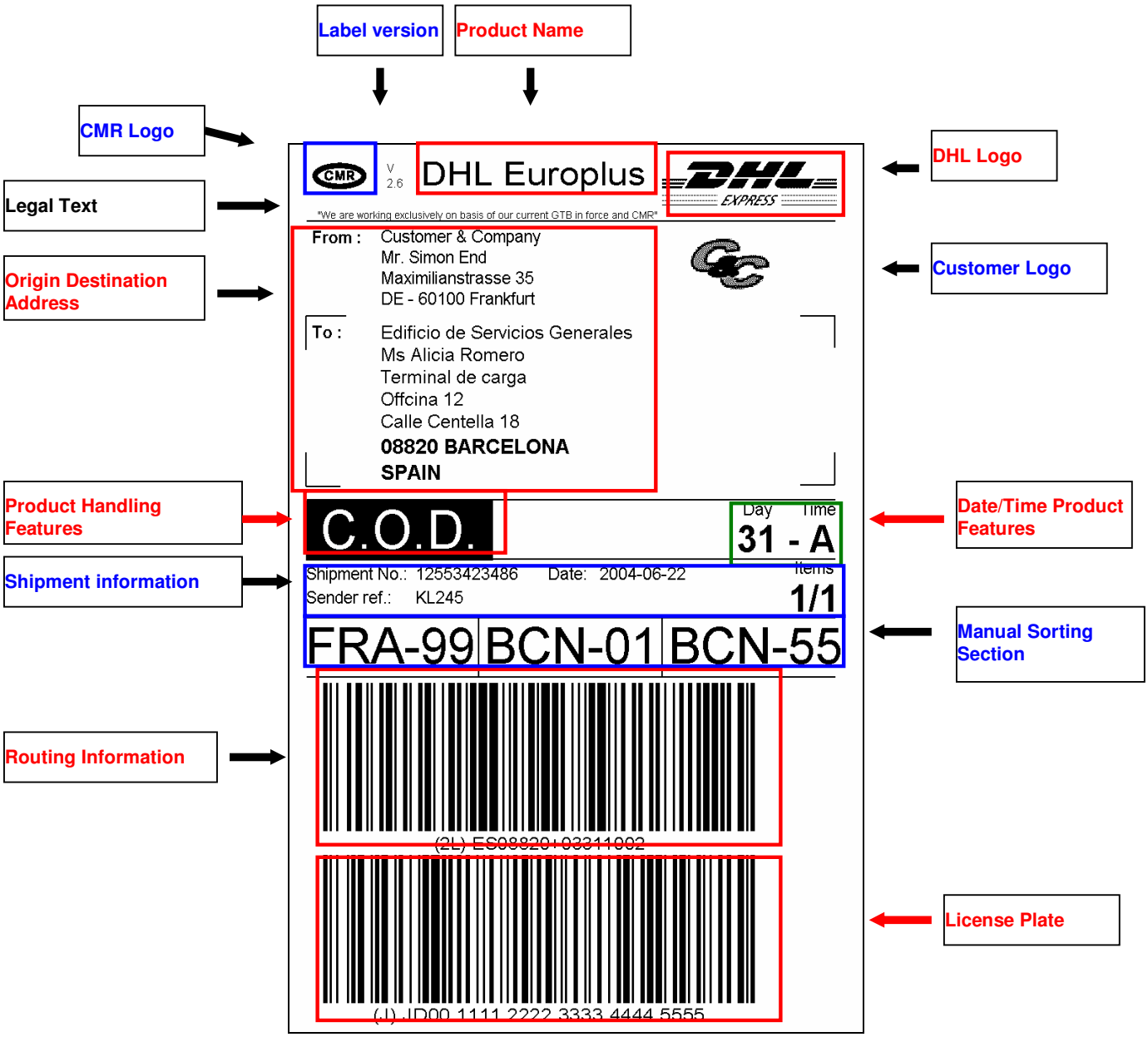
4.2 Layout

Below Label layout shows the 210 mm large format

4.4.4 **CMR logo**	4.4.2 Label version	4.4.3 Product	4.4.1 **DHL logo**
4.4.5 Legal Text			
4.4.6 Ship from address		4.4.7 Customer logo	
4.4.8 Ship to address			
4.4.9 Product features			
4.4.10 Shipment information			
4.4.11 Structured data file (2D Barcode)			
4.4.12 Product/region specific routing			
4.4.13 **Customs & general information**			
4.4.14 Customer segment			
4.4.15 Routing information			
4.4.16 License Plate			

4.3 Label example

Below A6 Label example displays the main sections:



4.4 Description of the label sections

4.4.1 DHL Logo (mandatory)

This section contains the carrier logo and is placed in the upper right corner of the label. Below the logo, information about the responsible DHL terminal or country organisation should be printed.

Any CAS system must print the logo as an image. Plain text is not allowed.
If it is not possible to insert images into the label, the logo text has to be printed in plain text instead. However, pre-printed logo's are preferred.

For transport labels, the black & white version of the logo's is sufficient. When using labels with a pre-printed logo, the official colour specifications for the yellow/red logo have to be followed.

The following logos shall be used: **DHL, DHL Express, DHL Danzas Air & Ocean, DHL Freight, DHL Solutions**, depending on the business unit dealing with the customer.

The **general "DHL" logo** is to be used wherever a customer does not want to change between logos when sending with different units.

Regardless of the kind of logo, the DHL terminal or country organisation could be printed below.

DHL ENTITY	DHL LOGO'S
General DHL Logo	
DHL Express Logo	EXPRESS
DHL Solutions Logo	SOLUTIONS
DHL Danzas Air & Ocean	DANZAS AIR & OCEAN
DHL Freight Logo	FREIGHT

4.4.2 Label version (optional)

This section depicts the version number of the label used. It helps verify whether the label version used abides by the latest specifications in use in terms of format and content. (e.g. : V 2.5)

The section is also intended to reflect the date of a destination routing table where applicable.

4.4.3 Product (mandatory)

This section contains the name of the service used (e.g. « DHL Europlus »).

4.4.4 CMR logo (conditional)

The CMR image is as follows :

If the printer does not support graphics, it is sufficient to print the letters CMR in bold font.

The CMR logo (Convention des Marchandises Routieres) is to be put on the top left side of the label and is used for all international road-based transport between many European countries.

The CMR legislation not only regulates the rights of the consignor (with whom the carrier has a contract), but also the consignee, with whom the carrier does not have a contract in most cases.

For **domestic shipments** the CMR logo is not mandatory, however it may be printed.

For **air-based shipments**, CMR is not relevant and this logo can be omitted.

For air-based shipments where CMR is not relevant, this section can be used to print the *relative number and total number of pieces in shipment*

4.4.5 Legal Text (optional)

The content of this section may vary significantly as it is subject to local legislations.

It may contain a reference to the General Terms of Business (GTB) . The recommended text is

"**We are working exclusively on the basis of our current GTB in force and CMR**".

Also the European legal text referring to the terms and conditions set out by the Warsaw Convention and DHL can be put in this field:

"***DHL standard terms and conditions apply. Warsaw Convention may also apply. Shipment may be carried via international stopping places which DHL deems appropriate***."

In the US, the **Commerce Control Statement** (**DCS**) may be used :

"**These commodities, technology or software were exported from the United States in accordance with the Export Administration regulations. Diversion contrary to U.S law prohibited**. **Shipment may be varied via intermediate stopping places which DHL deems appropriate**."

4.4.6 Ship from address (conditional)

This section can remain empty only if the piece already has a label with the "ship from" address that is compliant with the specifications mentioned above.

The "ship from" section refers to the address of the sender, for identification by the carrier as well as the receiver.

Under certain conditions an address other than the pick-up address may be printed here. Such an address, however, has to be a real address of the customer as well. No postal address should be used. Furthermore, certain transport products may require the actual pick-up address to appear here for customs reasons.

The word "From" is to be printed in the upper left corner of this section.

The address shall be structured as follows :

1	**Name of the sender / company name**
2	**Street and house number**, house number extension, special instructions like floor, room etc
3	**Contact name, phone number**
4	**Post code, town** (according to local standards – see above, maybe prefixed by ISO 3166 two-character country code)
5	**County or State** according to local standards
6	**Country name** (optional for domestic shipments, mandatory for cross-border shipments if ISO 3166 code is missing)

	For ***domestic* pieces** the country name may be printed in local language. For all ***cross-border* pieces** the country name has to be printed in English (may be followed by local translation).

The position of the **postcode** should be consistent with the standard layout for postal addresses in the country of the ship from address.

Typically, it will be one of the following four possibilities :

1	**preceding the town name** (e.g. France, Germany)
2	**following the town name** (e.g. USA, Malta)
3	**on a separate line at the end of the address block** (e.g. United Kingdom, Hungary)
4	**Non-existent** (e.g. Ireland)

For locations abroad, the **country name** should be written in its own line, as long as there is enough space within the maximum length of the label.

Other options are :

- Put the *ISO 3166 two character country code* before the postcode
- Add the full country name to the town line. In these cases a separate line containing the full country name is not required.

The word "**From**" may be written in a local language, however for international shipments the English word "From" must appear as well. In this case only English, or both languages (English and local language) may be used, separated by a "/".

For **air-based shipments**, the word "ORIGIN:" with the three or six character Origin Service Area Code is printed right of the section.

The maximum number of lines is restricted by the corresponding DHL-internal data flow and the maximum height of the block. Each line contains a maximum of **35** characters.

CAS

Any CAS application has to offer at least 5 lines for the consignee address

4.4.7 Customer logo (optional)

If the customer wants to put his logo onto the label, the logo has to be placed within this section. In this case the space the logo takes up will be taken out of the space for the from address. As the from address is printed in a smaller font than the to address, space should be sufficient.

4.4.8 Ship to address (mandatory)

This section is mandatory. If the piece already has a label with the "ship to" address that is compliant with the specifications mentioned above, the sentence "please refer to separate address label on piece" can be printed here instead.

4.4.8.1 Address

The word "**To**" may be written in local language, however for export shipments the English word "To" must appear as well. Alternatively, both languages (English and domestic) may be used, separated with a "/".

This section is **optional,** if the piece already has a label with the "ship to" address that is compliant with the specifications mentioned above.

The address shall be structured as follows :

1	**Name of the receiver / company name**
2	**Street and house number**, house number extension, special instructions like floor, room etc
3	**Contact name, phone number**
4	**Post code, town** (according to local standards – see above, maybe prefixed by ISO 3166 two-character country code)
5	**County or State** according to local standards
6	**Country name** (optional for domestic shipments, mandatory for cross-border shipments even if ISO 3166 code is present)

The position of the **postcode** should be consistent with the standard layout for postal addresses in the country of the ship to address.

Typically, it will be one of the following four possibilities :

1	**preceding the town name** (e.g. France, Germany)
2	**following the town name** (e.g. USA, Malta)
3	**on a separate line at the end of the address block** (e.g. United Kingdom, Hungary)

4	**Non-existent** (e.g. Ireland)

4.4.9 Product features (mandatory)

Product features encompass handling (printed left aligned) and date/time delivery (printed right aligned) features which are shown on the label to support DHL internal operational procedures. Only the product features that impact the delivery- or the routing/sorting process are shown on the label. The features to be used depend on the product chosen.

The remaining product features appear only on the accompanying documentation, such as the electronic information (EDI), the pickup- and delivery lists.

4.4.10 Shipment information (conditional)

The shipment section contains the following information.

4.4.10.1 The shipment number / waybill number (conditional)

This number is optional for *EUROPACK* and *EUROPREMIUM* shipments: If this information cannot be specified correctly for consolidated shipments, it must be left out. No dummy replacements with incorrect information are allowed.

The number is mandatory for all other DHL Express products.

For **air-based shipments**, the Destination Service Centre Code and the AIRWAYBILL number is printed in this section.

For **air-based shipments**, the AWB bar code must be printed in this section, positioned below the AWB number.

4.4.10.2 The customer number (optional)

This is a key to the database, referring to the party to be billed for the freight costs.

4.4.10.3 Sender's reference (optional)

This is a key to the sender's database, and can be used by the sender to reference and track pieces.

4.4.10.4 Shipment date (optional)

The dispatch date of the shipment / piece must be represented in ISO format (YYYY-MM-DD).

The entries "**Customer number**", "**Sender's reference**" and "**Shipment Date**" are optional. Each of them may be defined mandatory by any DHL Business Unit / region for own customers.

4.4.10.5 Relative number and total number of pieces in shipment (conditional)

The relative piece number of the piece is separated with a slash « / » from the total number of pieces. Example "**1/23**", "**2/23**", …

This information may be mandatory for some products.

Any one-piece product, such as ***EUROPACK***, has to carry the string "1/1".

The names of the pieces are listed in the following order:

Position No.	Piece
1 left	Shipment no.:
2 left	Customer number:
3 left	Sender's reference:

The entries "**Shipment no**." and "**Relative and total number of pieces**" are principally mandatory. However, they have to be suppressed if their correctness cannot be guaranteed.

4.4.11 Structured data file (2D Barcode) (optional)

This section contains an EDIFACT message, represented as a two dimensional (PDF417) barcode.

For domestic shipments, the 2D barcode can replace the routing information barcode

2D code can be used only where supported by local pick-up and never for direct injections!

4.4.12 Product/region specific routing information (optional)

This section contains domestic routing information that is specific to certain products and/or regions. In contrary to the routing barcode, it can contain destination information (delivery depot, tour, …) derived from destination postcode and product information. This information is used for manual sorting of pieces.

It must be omitted or left empty for pieces sent cross border unless it is preceded by the consignee country's ISO 3166 country code.

4.4.13 Customs & General Information (conditional)

The general information regarding the shipment, including description, weight and customs value is printed in this section on the label (Description, Service, Weight, Imp/Exp Type, Date, Customs Value, IV).

This section is only **mandatory** for **air-based shipments**

4.4.14 Customer segment (optional)

This section can be used by the sender to describe the goods that are sent.

There are no restrictions, as long as the information does not interfere with the other information on the label. If this section contains barcodes, the DHL Transport Label restrictions for other barcodes on the piece apply.

Please see also chapter (9.5.2)

4.4.15 Routing information (mandatory)

Please see chapter (5)

4.4.16 License Plate (mandatory)

Please see chapter (6)

5 Routing Information (mandatory)

5.1 General

DHL has decided to sort shipments and pieces through an ISO standard routing barcode.

The DHL Transport label is intended for use together with the detailed transport order data (e.g. through EDI). The routing barcode only contains the most relevant handling and delivery information, it does not replace the need for the full transport order data.

! No piece may carry more than one different routing barcode.

The DHL ISO routing barcode contains destination country, destination postcode, transport products and associated product features**.**

In general terms, it is recommended to print no routing label than a wrong one. When some routing information is not available or secure enough, the information is to be omitted but only when the information is not mandatory.

A clear distinction is to be made between cases where the available features are not chosen and cases where the requested information is not available or secure enough.

The general concept is as follows :

- ***Zeroes are indicated when "features are not chosen"***
- ***The field remains empty/blank when "information is not available or secure".***

In addition to this, below routing barcode structure table details which routing feature fields are mandatory, conditional or optional for both **data content** and **print out** level.

At data content level :

When a routing field is mandatory, it implies that the content is required to print the routing barcode.

When a routing field is **conditional**, it implies that the content is mandatory in some cases (e.g: for a specific product) and optional in other cases.

When a routing filed is optional, it implies that the content is not required in which cases zeroes are generally put.

In the context of the routing barcode, **optional** does not always imply that the routing field may remain empty. Some routing fields must always be populated at least with zeroes.

It is also assumed that if the data content is available for an optional routing field, the field will then be populated if appropriate and necessary for proper sorting purposes.

The **postcode and region specific fields** as well as the **+ separator** after the product/feature block are the only optional fields which must remain **empty** if the data is not available.

Please see also chapter on local temporary exceptions which details Air Express exceptions.

5.2 Routing barcode structure

The following two international ISO standards are supported when coding routing information in barcodes.

- The **ANS MH 10** routing barcode starts with the ANS MH 10 Data Identifier "**2L**" and is coded in Code 128.
- The **EAN.UCC** routing barcode starts with the EAN.UCC Application Identifier "**403**" and is coded in EAN 128.

While both standards are complementary and can co-exist without restriction (they do not differ in content but only in their technical structure), ***DHL will internally only produce ANS MH routing barcodes starting with (2L).***

The one-dimensional routing barcode may be omitted in some regions or for certain products.

Please refer to region-specific descriptions (5.8)

As mentioned above, a distinction has to be made between the information which will be captured and the information which will actually be printed out on the barcode. This distinction is mainly relevant for applications which need an indication on the output side.

Example : At **data content** level, the delivery date may not be available or required for the selected product, so the delivery date information will not be captured. The delivery date is optional as it must not always be captured.

However, at **print-out** level, the delivery date is mandatory for the following reason : even if the data is not captured, the delivery date field may not remain empty/blank, the field must be populated by *zeroes.*

The print-out level is either **mandatory** or **optional,** it means ***the whole barcode*** is either printed or not printed.

All routing information appears both in human and barcoded format except for the **data identifier brackets** which only appears in human readable format and not in the barcode.

Below table details **ANS MH** and **EAN** barcode routing structures

SECTION	ANS MH 10	EAN.UCC	DATA CONTENT	PRINT-OUT
OPEN BRACKET	(		Mandatory	Mandatory ***(only in human readable format)***
DATA IDENTIFIER	**"2L"**	**"403"**	Mandatory	Mandatory
CLOSED BRACKET	)		Mandatory	Mandatory ***(only in human readable format)***
ISO COUNTRY CODE	**2 alphanumeric digit**	**3 numeric digits**	Mandatory	Mandatory (if no data, do not print)
POSTCODE	Postcode of receiver address. **Variable length, max. 9 characters**; The postcode may never contain spaces or symbols **Remains empty in case of country with no postcode system.**		Optional (see also local temporary exceptions)	Optional May remain empty (see also local temporary exceptions)
FIELD SEPARATOR	"+" character		Mandatory	Mandatory (always put +)
PRODUCT CODE	**2 numeric digits (fixed length)**		Mandatory	Mandatory (if no data, do not print)

DELIVERY DATE	**2 numeric digits (fixed length)**		Optional	Mandatory (if no data, put 2 zeroes)
DELIVERY TIME	**1 numeric digit (fixed length)**		Optional	Mandatory (if no data, put 1 zero)
HANDLING FEATURE CODE(S)	**3 digit numeric number with leading zeroes (fixed length)** Sum of values of chosen product features		Optional	Mandatory (if no data, put 3 zeroes)
FIELD SEPARATOR	**FNC1** (Special Code 128 character, available in all three code sets.)	**"+"** character	**Conditional** To be used only if DHL product/feature block consists of less than 8 digits. See 4.10 for more info.	Optional May remain empty
LOCAL SPECIFIC CODES	Variable up to 6 numeric characters or up to 2 alphabetic characters, **for domestic use only**.		Optional	Optional May remain empty

Example of a routing code

(2L) BE3500 + 11311002123456

				Product/Feature block always consists of 8 digits				
(2L)	**BE**	**3500**	**+**	**11**	**31**	**1**	**002**	**12345 6**
Data Identifier	Iso Country Code	Postcode	Field separator	Product code	Delivery Date	Delivery Time	Product Feature	Region specific codes

5.3 Human readable/barcoded routing features

Most of the key routing information captured in the routing barcode and the human readable text underneath is also mentioned in other dedicated fields on the label (see also label example in chapter 3.2)

ROUTING SECTION	HUMAN READABLE LABEL FIELDS
ISO COUNTRY CODE	Appears in field “ship to address”
POSTCODE	Appears in field “ship to address”
PRODUCT CODE	Appears in field “product”
DELIVERY DATE	Appears in the right aligned field “delivery date”
DELIVERY TIME	Appears in right aligned field “delivery time”
HANDLING FEATURES	Appears in left aligned field “handling feature”
PRODUCT/REGION SPECIFIC INFORMATION	Appears centrally below handling feature section

5.4 Plausibility checks

DHL will not accept routing information barcodes that do not meet below requirements :

Any of the following linear barcodes will be considered the routing information barcode of the piece:

- Symbology Code 128 and starting with the ANS MH 10 Data Identifier **2L**
- Symbology EAN 128 and starting with the EAN Application Identifier **403**

In this case the following **plausibility checks** will be applied by DHL :

• The routing barcode shall only contain **numeric and upper case alphabetic characters** and separating plus ("+") characters.
• The two characters following the "2L" in the **ANS MH 10** routing barcode must equal a two **alpha ISO country code** according to ISO/IEC 3166.
• The three characters following the "403" in the **EAN** routing barcode must equal a three numeric ISO country code according to ISO/IEC 3166.
• The character string immediately following the plus "+" sign must contain at least 8 numeric only characters.
• This code may (although it is not mandatory) be followed by a separator (up to 6 numeric characters or up to 2 alphabetic characters, **for domestic use only**.)

5.5 Destination country code & postcode

The Data/Application Identifier is followed by the destination country code and postcode.

5.5.1 Country Code (mandatory)

Destination country is coded using ISO 3166 standard routing barcodes :

ANS MH 10	**Two** digit alphabetical routing codes
EAN	**Three** digit numeric routing codes.

5.5.2 Postcode (optional)

The postcode structures have to be converted, to produce postcodes **without** blanks, dashes or other symbols, to use in the routing barcode.

Given the postcode length vary from country to country and may also vary within the same country, this part of the routing barcode has to be terminated by a "+" (plus) sign.

The postcode part has to be omitted from the routing barcode when.

- **No postcode system is in place (e.g. : Ireland)**
- **No valid postcode is provided**
- **Destination postcode is missing**

Air Express exception

The following rule only applies to red DHL products and applications.

The blank/missing postcode part in the barcode must be replaced by a DHL valid 3-letter destination code, preceded by a colon ":", e.g. for Ireland "**:ORK**" will need to be used for Service Area "Cork". Additional example would be 2LLB:BEY+42101073)

See also section on local temporary exceptions

Postcode table excerpt:

Country	Structure	Conversion
NL	9999 AA	9999AA
PL	99-999	99999
GB	AAA 9AA	AAA9AA
SE	999 99	99999

(Note : “9” represents a number, “A” represents a capital letter.

The complete and most up-to-date list of all postcode formats is downloadable from the following location :

http://www.lhr-sys.dhl.com/OperationsBPA/pci/html/postcode_formats.html

5.6 Products (mandatory)

Each DHL product is allocated **a two-digit product code** which has to follow the "+" sign.

Product	Code
EUROPACK	01
EUROPREMIUM	02
…	…

5.7 Product Features (optional)

5.7.1 General

DHL product portfolio consists of relatively few products each of which can be completed by product features.

Not all transport order data supplied by the customer must appear on the label. While the product must always be coded, only product features that may be relevant for sorting and delivery processes are shown on the label.

The product feature section contains :

- **Date/Time Delivery Information**: supports delivery process on a definite date and time.
- **Handling information** : supports both sorting and delivery processes

The label does not replace the need for the transport order data but leads the piece through the network without the need of the corresponding order data.

Like most of the information on the label, the product features are represented in human readable as well as bar-coded form to support both manual and automated processes.

The format for feature coding shall be the same for all products but the meaning may differ across the diversity of products.

While most of the product features are combinable (e.g. COD with date certain delivery), some features are mutually exclusive - such as Hold at Depot, Delivery on appointment and Date Certain Delivery).

5.7.2 Delivery Date (optional)

On the right side of the product features section, the delivery date or time features are shown.

The delivery day indicated on the label does not represent the calculated day of delivery based on destination and product. It is only used for deliveries that are ordered for a specific calendar date

The specified day of the month is to be represented in 2 digit numeric format with leading zeroes.

The month or the year is not shown although the day alone could be ambiguous. The reason being that the information on the label is only supportive and does not have to contain the whole order data.

This field may also contain non-numeric values, representing delivery instructions that cannot be combined with a certain delivery date.

Examples:

DELIVERY INSTRUCTION	LABEL FORMAT
"Saturday delivery"	“S”
"Delivery on Appointment"	“A”

The barcode structure allows for 99 different delivery date codes per product. Date codes can have different meanings for different products. The delivery date part of the routing barcode consists of two numeric digits always.

Example:

Delivery date codes	Product	Value	Handling information
No fixed date delivery	All	00	(none)
Fixed day delivery (day of the month)	…	01-31	01-31
Delivery on appointment	…	51	A
Saturday only delivery	…	50	S
Hold at depot (no delivery)	…	52	H
	...	...	...

5.7.3 Delivery Time (optional)

If a time window is chosen, the code corresponding to the time window is shown.

In case both day and time window features are mentioned, the corresponding codes should be separated by a minus sign ("-").

Example: "05-X". The separator is only required when both features are chosen.

The delivery date/time information has a header, consisting of the words "day" and "time". The words "day" and "time" may be written in local language, however for international shipments the English words "day" and "time" must appear as well. In this case only English, or both languages (English and domestic) may be used, separated with a "/".

The barcode structure allows for 9 different time codes per product. I.e., time code 1 could have a different meaning for different products. The delivery time part of the routing barcode consists of one numeric digit.

Example:

Time Feature Text	Time Code	Handling Information
No fixed delivery time	0	(none)
Pre 09:00	1	A
Pre 12:00	2	B

5.7.4 Handling information (mandatory)

The product feature part of the routing barcode is always **a three digit numeric number** based on a binary. The values of the chosen product features are added up (with leading zeroes).

Example :

The routing code is : (2L) ES08820+**02**312**003**

02 shows the product code

003 shows the product features used (it is the sum of C & C.O.D. product features (**001 + 002**))

This binary system enables to combine features without increasing the barcode length and allows to check for features set or to check for combination of features that are not allowed.

The product features not officially defined as handling relevant are not shown on the label. These product features appear only on the accompanying documentation, such as the electronic information (EDI), the pickup- and delivery lists.

The following product features are shown in the handling information part

Product feature codes	Value	Handling information
No Product Feature	000	(none)
Customs clearance	001	C
Cash on delivery	002	COD (or C.O.D.)
Receiver Pays	004	EXW
Return piece	008	RETURN
…	016	…
…	032	…
…	064	…
…	128	…
…	256	…

When more than one handling feature is mentioned on the label, these are separated by a minus sign ("-").

Example: "**COD-EXW**".

When none of the above features is chosen, the left side of the product features section remains empty.

For customs clearance relevant pieces a "C" is shown. All pieces shipped across a customs boundary are considered customs relevant pieces. However, bonded goods do not need to be marked with a "C".

5.8 Product/Region specific routing information (optional)

This section can be skipped in regions without specific domestic routing information.

The region specific routing information may only be used domestically, or on international shipments where it is filled with the routing information required by the destination region, as bilaterally agreed before. For other shipments it may not be used.

- Using the ANS MH 10 routing barcode, this segment has to be separated from the standard routing code content by an FNC1 character, available in all three Code 128 specific code sets. The scanner passes this through as "]C1".

- Using the EAN.UCC routing barcode, this segment has to be separated from the standard routing code content by a "+" sign.

Example :

	DI	Country	Postcode		Product	Date	Time	Feature	Domestic routing code
FNC1	**403**	**276**	**81541**	**+**	**01**	**00**	**0**	**000**	**123456**
Start character		Germany			EUROPACK	No fixed date	No fixed time	No features	e. g. Street and house no.

Example :

	DI	Country	Postcode		Product	Date	Time		Domestic routing code
FNC1	**403**	**276**	**81541**	**+**	**02**	**00**	**0**	**+**	**123456**
Start character		Germany			EUROPREMIUM	No fixed date	No fixed time		e. g. Street and house no.

! The structure of the domestic part is country-specific, a general structure cannot be described here.

5.9 Routing barcode examples

5.9.1 ANS MH 10

- **EUROPACK** piece, without any features but with region specific routing information:

	Country	Postcode		Product	Date	Time	Feature		Domestic routing code
2L	**DE**	**81541**	**+**	**01**	**00**	**0**	**000**	**FNC 1**	**556677**
	Germany			EUROPACK	No fixed date	No fixed time	No features	Separ-ator	Street and house number

- **EUROPREMIUM** shipment, to be held at depot, with customs clearance:

	Country	Postcode		Product	Date	Time	Feature
2L	**CH**	**3000**	**+**	**02**	**52**	**0**	**001**
	Switzerland			EURO PREMIUM	Hold at depot	No fixed time	Customs Clearance

(Future features) a EUROPREMIUM shipment, to be delivered on the 10th, in time code 1, with customs clearance and the future product features with codes 008 and 064:

	Country	Postcode		Product	Date	Time	Feature
2L	**NL**	**1023FG**	**+**	**02**	**10**	**1**	**073**
	The Netherlands			EURO PREMIUM	Delivery at 10th of month	Time code 1	73=1+8+64

- EUROPACK piece, without postcode **(only for countries with no postcode system)** and without any features:

	Country		Product	Date	Time	Feature
2L	**IE**	**+**	**01**	**00**	**0**	**000**
	Ireland		EUROPACK	No fixed date	No fixed time	No features

5.9.2 EAN

- EUROPACK piece, without any features but with region specific routing information:

	AI	Country	Postcode		Product	Date	Time	Feature	Domestic routing code
FNC1	**403**	**276**	**81541**	**+**	**01**	**00**	**0**	**000**	**123456**
Start character		Germany			EUROPACK	No fixed date	No fixed time	No features	e. g. Street and house no.

Remark : In this example is no separator between features and domestic routing code, because the product and feature code are 8 digits, if less than 8 digits it needs a separator.

- EUROPACK piece, without postcode **(only for countries with no postcode system)** and without any features :

	AI	Country		Product	Date	Time	Feature
FNC1	**403**	**372**	**+**	**01**	**00**	**0**	**000**
Start character		Ireland		EUROPACK	No fixed date	No fixed time	No features

- EUROPREMIUM shipment, to be held at depot, with customs clearance:

	AI	Country	Postcode		Product	Date	Time	Feature
FNC1	**403**	**756**	**3000**	**+**	**02**	**52**	**0**	**001**
Start character		Switzerland			EURO PREMIUM	Hold at depot	No fixed time	Customs Clearance

5.10 DHL Additional Routing labels

5.10.1 Additional routing label

The production of a separate routing label is mainly for DHL internal use.

The additional routing label has to be attached to the original label by DHL whenever the existing label does not contain a (proper) routing code.

Such cases occur when :

1	a manual (pre-printed) Label is used by the customer
2	a parcel needs to be re-directed, e. g. in case of a routing error or a return to sender.
3	A DHL unit decides to add unit specific routing information to import parcels.
4	No routing barcode had been printed by the customer..

5.10.2 New routing label

The following steps take place whenever a new routing label needs to replace an existing one :

1	The old one has to be made unreadable, preferably by **a thick vertical(!) line** across the old barcode.
2	Alternatively, the old barcode can be over-labelled by the new one, provided no other relevant information becomes unreadable.
3	In case of returns, sender and consignee addresses change as well. Thus it is recommended to replace the whole label by a new one in case of returns. The software that creates those labels could do that automatically on the basis of the PAN.

The necessity to attach a separate RETURN sticker remains unchanged.

5.10.3 Label Layout

Below An ANS Code 128 example of a (2L) routing label with all fields is shown below:

5.10.3.1 Mandatory/optional label sections

The routing label consists of the following segments of the standard automated label format :

1	**DHL Logo**	**Mandatory**
2	**Label specification version**, including : Version number Date & Time of label printout User ID	**Optional**
3	**Product information**	**Mandatory**
4	**CMR logo**	**Conditional**
5	**Human readable routing information**	**Mandatory**
6	**Shipment information**.	**Optional**
7	**DHL specific routing information**.	**Optional**

8	**Routing barcode**	**Mandatory**
9	**License Plate in human-readable form** (enabling a positive proof of routing label belonging to the right piece in case of doubt)	**Optional**
10	**Location Information** where the label has been printed + printing date + gate of printer, ….	**Optional**
11	**Features** that have been relevant for the first delivery attempt (such as Saturday delivery or Payment), do not have to be re-printed in case of a return. As of today, the only relevant feature that is valid for returns also, is the "C" for customs clearance	**Optional**
12	**Shipment information** could be added in case of multi-piece shipments, whenever the piece does not contain any shipment information (e.g. in case of manual (pre-printed) labels). For the label width, the same rules apply that are valid for full automated labels.	**Optional**

5.11 Label certification

Please consider that any label has to be checked and approved by the **DHL Label Certification Office**.

6 License Plate (mandatory)

6.1 Concept

The International Standard **ISO/IEC 15459** has been created to allow unique identification of transport units, independent of the industrial sector for which the unit is transported. In this system each transport unit is assigned a number (called License Plate Number) which distinguishes it unambiguously from all other transport units.

All transport units are handled by several parties in the transport chain – shipper, consignee and one or more carriers, customs etc... Each of these parties should be able to uniquely identify the unit so that reference can be made to associated information such as address, shipment number, weight, etc…

The License Plate is a unique piece identifier which uniquely identifies each single transport unit at lowest level, commonly named "piece level".

Each transport unit must be equipped with a DHL Transport label containing a License Plate.While DHL recommends to use ANS MH 10 standards, DHL will also accept and handle other License Plate standards as long as they comply with the ISO standards referred to in this document.

Content wise, DHL will always treat customer License Plates as **non-speaking** identifiers.

ISO has developed the License Plate (LP) to identify a "Transport Unit" which means the same as "piece" in DHL terms. The License Plate identifies one piece and must remain visible on each piece both in barcoded and human readable form at the bottom of each label.

! No piece may carry more than one License Plate.

Each piece may belong to one or many shipments – ***but to not more than one at a time!***

While the shipment number is a *process* identifier which changes whenever the direction of a piece changes - returned or forwarded shipments build a new transport process, the License Plate number remains unchanged for the whole lifetime of a Transport Unit.

6.2 Requirements

A unique piece identifier or License Plate number must :

1	begin with a combination of characters, the issuing agency code (IAC), which is assigned by the Registration Authority to the issuing agency. The EAN organisation issues a License Plate (called the Serial Shipping Container Code, SSCC), with numeric Issuing Agency code. Other issuing agency codes usually contain letters.
2	be constructed according to the specification of the issuing agency
3	be unique so that no user can re-use a number within a sufficiently long time period so that the first number has a unique meaning for every user of this standard. DHL has decided to set this period to 12 months; recommended for a longer time e.g. 3 years
4	only contain numbers or upper case letters (no lower case letters or other characters). This means only the characters "0".."9" and "A".."Z" are allowed; drawn from ISO/IEC 646
5	not contain more than **35** characters for reasons of space. Because no extra information other than the unique number is expected in the barcode, in practice the barcode will not likely contain more than 20 characters.

CAS

Although, according to the official standard, both Code 128 and Code 39 are allowed to encode License Plate numbers, any barcode produced by a CAS system or any DHL internal software has to use Code 128.

6.3 Data/Application Identifier

The data identifier is an ISO string of characters which identifies the content of the data which follow.

All DHL issued Licence Plates will use the Data Identifier “J”

All DHL Licence Plates LPs begin with “JD00“ or “JD01” followed by a space.

For readability reasons it is strongly recommended to group the code by blocks of four digits, beginning with the first character of the License Plate.

It is mandatory for **air-based shipments** to print out of License Plate in groups of four digits. Recommended also for all other products.

example: **(J) JD01 2012 3456 0123 456**

A collection, consisting of pieces marked with License Plate identifiers, must not be marked with a License Plate identifier, unless the labels of the contained pieces are covered (e.g. by black shrink foil).

The data identifier preceding the License Plate is NOT part of the License Plate. It does not have to be considered when determining the License Plate length. Two License Plates are identical if their only difference is a deviating data identifier (e.g. “J” versus “2J”). However, It does have to be printed on the label both in human readable and barcoded format but the barcoded format should not include the brackets. These parentheses are used for clarity only.

(The barcoded form of the data identifier and licence plate is then “JJD01…….”. & the human readable form should be in the form “(J)JD01 …..”.)

6.4 Issuing Agency Code

The standard ISO/IEC 15459-1 is based, amongst others, on the operation of unambiguously identified Issuing Agencies.

Each of these agencies is identified by an Issuing Agency Code (IAC), which is assigned and registered by the ISO/IEC 15459 Registration Authority, the Nederlands Normalisatie-instituut (briefly called NEN).

The Issuing Agency Code immediately follows the data/application identifier and consists of a two or three alpha characters for ANS MH 10 standards or a numerical value (from 0 to 9) for EAN License Plates.

DHL will only accept License Plates which are compliant with the rules settled by the recognised issuing agencies as per ISO/IEC 15459-1.

Please see the complete list of IAC codes in chapter **(16)**

6.5 ANS MH & EAN Standards

The ISO License Plate recognises two sub-standards : **ANS MH 10** and **EAN SSCC**.

6.5.1 ANS MH 10

Any linear barcode using barcode symbology CODE 128 or Code 39 and starting with the ANS MH 10 Data Identifier J, 1J, 2J, 3J,4J, 5J or 6J will be considered the (ANSI) License Plate of the piece.

The License Plate has the following content structure :

- **Data Identifier** : indicates that an LP follows but is not part of the LP
- **Issuing Agency Code :** defines a unique number allocated to the Issuing Agency by the Registration Authority.
- **Format identifier :** defines the organizational unit responsible for the structuring of the information following the Format identifier.
- **Allocation Domain :** defines the organizational unit or domain responsible for the uniqueness of the serial number.
- **Serial reference number :** is allocated by the organizational unit defined in the Allocation Domain or Format Identifier.

  (this unique part is determined by the owner and is unique for a period of at least 12 months. Recommended for a longer time e.g. 3 years).

Below table displays mandatory fields of the LP

Content	**Part of LP?**	**Mandatory /Optional**	**Length**
Data Identifier	No	Optional	Fixed, 1 digit
Issuing Agency Code	Yes	Mandatory	Variable, alphanumeric
Format Identifier	Yes	Mandatory	Fixed, 3digits
Allocation Domain	Yes	Mandatory	Variable, numeric only

Serial Number	Yes	Mandatory	Variable, numeric only
Check Digit	No	Optional	Fixed, 1 digit

In this case the following plausibility checks will be applied by DHL:

- The License Plate must not contain more than **35 characters**
- The License Plate may only contain numeric and upper case alphabetic characters drawn from ISO/IEC 646 – they must not include any lower case character or punctuation mark.

The License Plate must start with a string of characters representing an issuing agency assigned by ISO. This plausibility check is applied whenever access to a table of current issuing agency codes is available (the list of Issuing Agency Codes is not subject to frequent change).:

All Issuing Agencies with exception of EAN are obliged to make use of the ANSI classification.

Example of ANS MH 10 LP

(J)	**JD00**	**123**	**3000**	**1234567**
Data Identifier **(not part of LP)**	Issuing Agency Code	Format Identifier	Allocation Domain	Piece Serial Number

6.5.2 EAN

Any linear barcode using barcode type EAN128 and starting with the special character FNC1 and EAN Application Identifier 00 will be considered the (EAN) License Plate (Serial Shipping Container Code) of the piece.

The EAN SSCC (Serial Shipping Container Code) has the following content structure :

- **Application identifier** : indicates that an LP follows but is not part of the LP
- **Extension digit** : is assigned by the user according to internal needs
- **International Location Number (ILN)** : company prefix allocated to an UCC/EAN system user by an EAN numbering Organisation or the UCC.
- **Serial Number** : is assigned by the user and is structured according to internal needs;
- **Check digit** : is calculated according to the UCC/EAN algorithm

the following plausibility checks will be applied by DHL:

- The License Plate must contain 18 characters (not counting the “00” application identifier).
- The License Plate shall only contain numeric characters.

The barcode is completely numeric and (including the Application Identifier and the check digit). In the readable text below the barcode the Application Identifier is always put between parentheses.

Example of EAN SSCC

(00)	**3**	**4012345**	**123456789**	**5**
Application Identifier **(not part of LP)**	Extension Digit	ILN	Serial Number	Check Digit

Please see appendix for more examples of License Plate structures.

7 Barcode requirements

7.1 General

As a general back-up routine, all barcoded data content must also be translated into human readable (plain text) format to remedy cases where the barcode is not readable.

7.2 Non-DHL barcodes on the piece

DHL recommends to use **Code 128** or **EAN 128**.

DHL will also accept the Code 39 for ANS coded information although it is not the recommended barcode symbology.

The DHL Transport Label may require a number of barcodes needed by DHL:

- **The License Plate** This linear barcode is mandatory on all pieces.
- **The routing barcode** . This linear barcode is mandatory for all international shipments, as well as for domestic shipments that do not contain a 2D-barcode.
- **The 2D barcode.** This barcode is optional.

The customer can apply more barcodes to the piece, linear as well as 2-dimensional, for his own purposes. However, care must be taken that these other barcodes do not interfere with DHL internal processes.

For this reason, the following restrictions apply to other barcodes on the piece:

7.3 Linear barcode symbology

DHL uses Code 128 as standard barcode symbology. For bar-coded information defined under EAN-standards (such as the EAN SSCC) the EAN 128 barcode symbology will be used.

According to License Plate standards, DHL will also accept the Code 39 for ANSI coded information. This barcode symbology however is not the preferred barcode symbology.

Code 39 must not be used on any labels produced by DHL or systems created by DHL.

CAS

When Code 39 is used, special care has to be taken that the barcodes do not exceed the maximum length requirements, due to the low information density of this barcode.

Please see chapter **(7)** for further info on barcode symbologies

The following requirements are imposed on the one-dimensional barcodes on the DHL Transport Label. The use of the minimum values of the specifications is recommended.

Symbol height	The minimum height for the barcode is 25 mm. However, a barcode height of at least 28 millimetres is recommended
x-dimension	The minimum narrow element dimension (x-dimension) shall not be less than **0,33 mm** . The x-dimension shall not exceed **0,51 mm**. **Recommended 0,50 mm.** It is recommended to use the biggest x-width within the possibilities of the printing device used.
Quiet zone	Linear barcode symbols must be printed with leading and trailing **quiet zones not less than 5 mm**. Quiet zones are not part of the barcode itself, i.e. a barcode of 91 mm needs at least 101 mm space on the label.
Quality	The linear barcode quality must, as a minimum, conform to **"Grade B",** tested according to EN 1635, Test Specifications for Bar Code Symbols.
Orientation	The code must be of the fence type (standing).
Barcode width	The width of a barcode depends on the following parameters: **The x-dimension** **The total number of characters that needs to be represented**. **The proportion of alphanumerical characters that needs to be represented**. Since Code128 supports three character sets (A, B and C) it is possible, using character set C, to represent 2 numerical digits with one single barcode symbol. Strings containing mainly numerical digits will therefore be shorter than strings containing mainly alphanumerical characters. **The distribution of the alphanumerical characters in the string of characters**. Within the barcode, it is possible to switch from one character set to another, using a switch symbol. However, this switch symbol is also a barcode symbol in itself that uses up space.

	Therefore the string “ABCDEF123456” for example would be shorter to represent than “AB12CD23EF56”.
Translation of the Code Content	To facilitate a backup routine, if the barcode is not readable, the data content in the barcode symbol must be written in plain text immediately below the symbol

7.4 Two dimensional barcode symbology

DHL uses the PDF417 code to record shipment data belonging to the associated shipping unit. Among the three PDF types. DHL supports the standard PDF.

DHL has chosen for the use of an X-dimension of 10 mil or a multiple of it.

The minimum narrow element dimension (x-dimension) shall not be less than **0,1mm**.

The relationship between height and width of a single stripe or white space is called the ‘**aspect ratio**’. For the PDF417 a standard value of **1:2** is used. This means that the height is twice the width.

7.4.1 Message encoding

Encoding shall be done according to ***ISO 15434*** Transfer Syntax standard, using format indicator 02 (indicating the format "Complete EDI message / transaction"). That means that the whole message is encoded rather than replacing the UN/EDIFACT segmenting by Transfer Syntax format headers. As a result, the whole message can easily be encoded in PDF417 following this standard and can subsequently be directly passed on to an EDIFACT translator after scanning.

7.4.2 Message coding

The EDIFACT message is to be coded into a PDF 417 barcode according to PDF 417 standard ***ISO 15438.***

7.4.3 Error correction

PDF417 symbols can be damaged and still decoded.

The amount of damage that a symbol can withstand is its error correction capacity. DHL uses **level 5**.

Please see chapter **(7)** for further info on two dimensional barcode symbologies.

7.5 Plausibility Checks

DHL will apply the following plausibility checks to the piece identification barcode (this means, DHL will not accept License Plates that do not meet the following requirements) :

- **On any piece, there is to be only one License Plate**
- **For the time being, the License Plate may not be identical to a License Plate that has already been processed by DHL within one year**.

7.6 Check Digits

Error detection codes (also referred to as check digits) are available at two levels:

- **Barcode internal level** (coded as a character in the barcode, before the stop sign, but not visible in the human readable text.)
- **Piece identifier level** (coded in the identcode, visible in human readable text.)

The check digit at internal barcode level is inherent to the code 128.

The check digit at piece level, however, is not defined within the ISO License Plate standard. It can be defined by each Issuing Agency or user.

DHL supports the following checksum methods for both linear and two dimensional barcodes :

- linear barcode symbology : check-digit in the code 128
- two-dimensional barcode : the error-correction possibilities of the PDF417.

Please see also chapter **(15.4)** for further info on checksum methods

8 DHL Label Dimensions & Formats

8.1 Fonts and sizes

Although **ARIAL** font is recommended other fonts may be accepted if they provide an equal legibility

The following values specify ***minimum font sizes***. However, the maximum height per block has to be considered. Furthermore, the font size for the consignee address must be bigger than the font size for the sender address. For each address, the postcode and town font size must not be smaller than the size of the biggest font used in the respective address.

<table>
<tr><th>DHL LOGO</th><th>VERSION NUMBER</th><th>PRODUCT</th><th>CMR</th></tr>
<tr><td>Minimum: 10 mm high and 30 mm wide.
alternatively printed in plain text,
min. 10 mm high, bold black font.</td><td>1 mm min. high black font.</td><td>5 mm high bold black lettering.</td><td>alternatively text "CMR“
Min. 2 mm high, bold black font. Only where applicable.</td></tr>
<tr><td></td><td colspan="3">GTB REFERENCE
1 mm min. high black font.</td></tr>
<tr><td colspan="3">SHIP FROM ADDRESS
"FROM" - 2 mm high black font.
Senders Information - Min. 2 mm high black lettering, (7 lines max.)</td><td>CUSTOMER LOGO
must not exceed 25 mm in height</td></tr>
<tr><td colspan="4">SHIP TO ADDRESS
In the top left corner of this section the word “To” is printed. The ship to address has hooks, i.e. thicker corners to highlight it from the other information blocks. The recommended size of the hooks is 4 x 4 mm and the line thickness is > 0.5 mm.
"TO" – 2 mm high black font.
Delivery address - Min. 2,5 mm high black lettering,
must not exceed 30 mm in height
- Post code, Town (maybe prefixed by ISO 3166 two-character country code), 3 mm high bold lettering
- Country name 3 mm high bold lettering
(7 lines max.)</td></tr>
<tr><td colspan="4">PRODUCT/FEATURE CODES : 26pt Font size, 7.06 mm Font Height, 56 dots at 200 dpi (BOLD)
Due to the importance of the handling information for the efficient routing of the pieces through the network, this information must be printed in reverse video or – should the printing hardware not support reverse video – with a surrounding frame with the header “Handling” information printed left aligned, to increase visibility.</td></tr>
</table>

SHIPMENT INFORMATION
SHIPMENT IDENTIFIER **6 mm** min. high bold font, in white characters on a black background, must not exceed **20 mm** in height the inverse block must be at least **10 mm** wide, **1 mm** high black font "Day" and "Time" mandatory if feature used. **1 mm** high black font "Handling" (right above in the frame), mandatory if a frame used instead of inverse printing **CUSTOMER NUMBER : 2 mm** high black font **SENDER'S REFERENCE : 2 mm** high black font **SHIPMENT DATE : 2 mm** high black font **RELATIVE NUMBER AND TOTAL NUMBER OF PIECES IN SHIPMENT (BOLD)** **4,5 mm** high bold black font, must not exceed **20 mm** in height
2D BARCODE (PDF 417)
DHL INTERNAL ROUTING INFORMATION The routing barcode must not exceed a length of **91 mm** **8pt** Font Size, **2.82 mm** Font Height, **22 dots** at 200 dpi
LICENSE PLATE **8pt** Font Size, **2.82 mm** Font Height, **22 dots** at 200 dpi minimum quiet zones left and right : **5 mm** each minimum barcode height : **25 mm** barcode must not be longer than **91 mm** minimum x-dimension :l not be less than **0.33 mm** and not more than **0.51 mm** Content of the Barcode must be printed in **2mm** black plain text below the barcode

"Font height in mm" and "Size in dots at 200 dpi" are rounded values only to make programming easier.

8.2 Label orientation

The default and preferred orientation for the DHL label is the **portrait orientation** A landscape format is only allowed as an exception, to be approved by the *DHL Central Label Administration Office*.

The label width must be **within 95 and 105 mm**.

The preferred label length is **148 mm**, but it can be longer if necessary. A general maximum length is not defined. However, each customer must be given the possibility to print labels that are not longer than 250 mm.

As an exception, the label can be divided over two smaller sized labels. In this case the label that does not contain the unique piece identifier in bar-coded form, must contain the unique piece identifier at least in human readable form (minimal font size 8 point).

CAS

For any CAS system, the portrait format is **mandatory**.

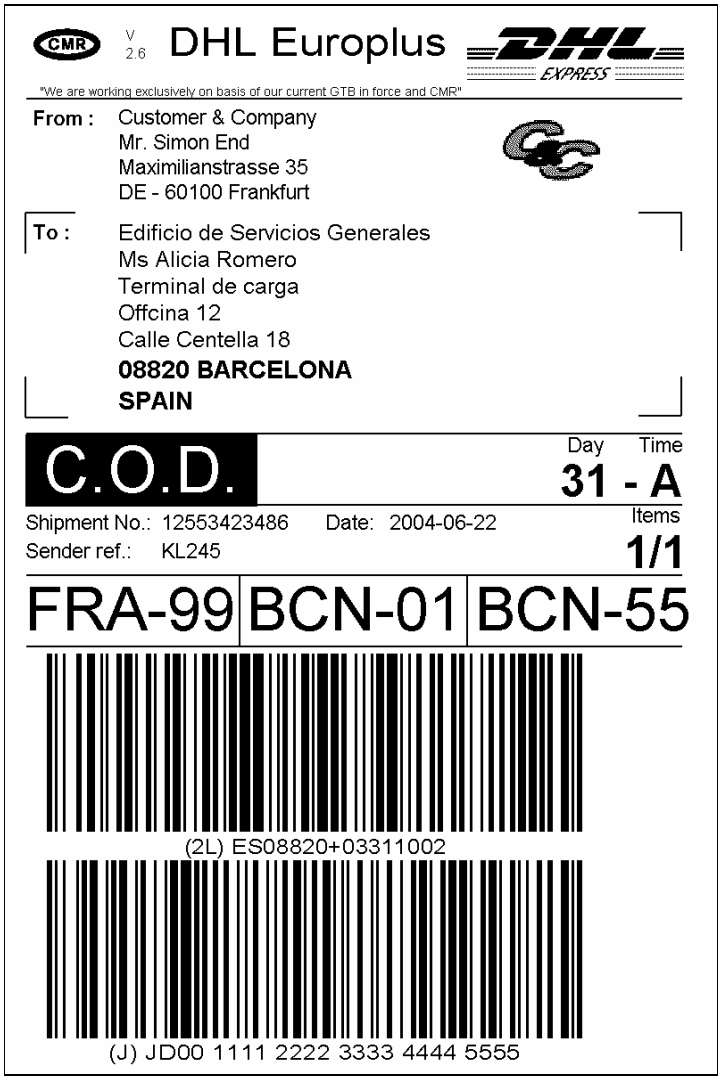
9 Label Examples

This chapter displays examples of the major different possible DHL Transport Label layouts. Dimensions mentioned below are recommended indications and may be subject to allowed variations as per specifications described in chapter **(8)**

9.1 A6 Layout

9.1.1 A6 Layout with all sections

This A6 DHL Transport Label shows all possible **mandatory**, **conditional** and **optional** sections. (**Length : 148 mm Width : 100mm)**



9.1.2 A6 layout with only mandatory sections

Below label only shows the minimum required information needed on an A6 DHL Transport Label(**Length : 148 mm Width : 100mm)**

9.2 Large Label Layout

This large label also contains a 2D Barcode. (**Length : 210 mm Width : 100 mm**)

9.3 A5 Layout

Each standard A6 DHL Transport Label can be added by an A6 customer segment which creates an A5 Label. See below example :

Length : 148 mm

Width : 200 mm

Please see also chapter **(9.5.2.1)** for an A5 Network Label example.

9.4 A7 Layout

9.4.1 Small DHL Label Layout

When the piece already has a label with the “ship from” and “ship to” addresses which is compliant with the present specifications, below small label can be produced.

Length : 100 mm

Width : 100 mm

9.4.2 A7 Routing Label layout

Below separate routing label format is mainly produced by DHL when the routing barcode on the original label is either absent, insecure or wrong.

Length : 70 mm

Width : 100 mm

9.5 Air-based shipment labels

9.5.1 Air-based Network Labels

The DHL Network labels are produced on ***DHL premises*** and are used to prepare Flyer, Express Envelope, Parcel and Non-Conveyable material for transfer through the DHL & DPWN network.

Information captured at 'pick-up' on a Courier Data Capture Device or keyboard manual entry is used to produce the label. The base label is white with the option of pre-printing the DHL Express logo in DHL red on Postyellow in the top left-hand corner, or bitmapped in black. To ensure network clarity and ease of recognition the labels follow strict content, format and layout criteria, which are indicated in the following sections.

Complete adherence to the bar code specification is mandatory as poorly produced or faulty bar codes significantly increase the risk of service delays.

The main components of the Network labels are:

1. DHL Logo
2. Product Code
3. Piece Number and Number of Pieces
4. CMR Logo (REX only)
5. Ship to address
6. Origin Code
7. Product Features
8. Delivery Time Window Details
9. Shipment identifier
10. Sender's reference
11. Destination Service Area Code
12. Sort Code
13. Air waybill Number
14. Air waybill Bar Code
15. Routing Information Bar Code
16. Routing Information
17. Licence Plate Bar Code
18. Licence Plate Identifier

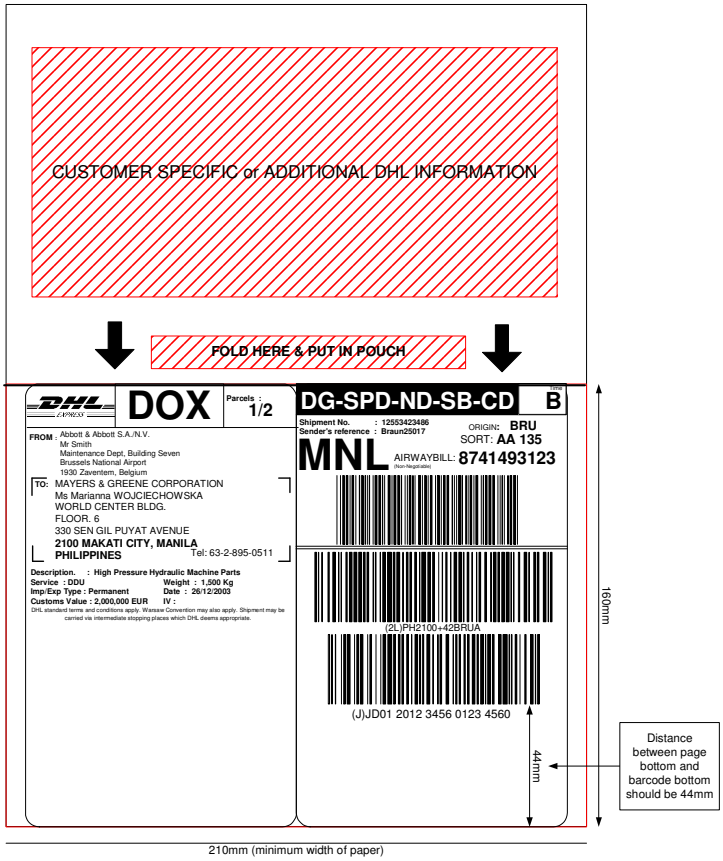
9.5.2 A5 eCOM Labels

The eCOM label is an Air-based shipment label which is normally printed on **A5 format** and have the same function as other eCOM labels and is produced on the ***customers' site***. It is generally printed on part of a larger sheet by a laser printer. The larger sheet is folded and placed in the pouch of a flyer or express envelope. For other material it is placed in a self-adhesive pouch and attached to the piece.

The bar codes must be correctly positioned in the pouch to ensure that the bar codes can be read properly and the material be processed effectively. Some areas of the pouch impact the readability of the bar codes and the bar codes must be kept out of these areas. For this reason, the bar codes must be located as shown on the larger sheet, keeping to the dimensions indicated on the drawing below, and the sheet must be folded where indicated.

The main components of the E-com Network labels are :

9.5.2.1 Example of eCom label – A5 format



Please see also chapter **(5.10).**

10 Label usage

10.1 Label placement

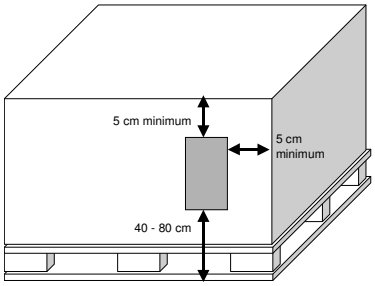
Labels should be fixed at a suitable location where there is a minimal risk of damage. The surface where the label is placed on should be even and the label should stick on it durably.

10.1.1 Unit loads (pallets)

The following requirements should be followed :

1	Labels should be placed on the side of the piece with the human readable information parallel to the natural bottom of the piece.
2	Each pallet must have exactly one **DHL Transport Label**.
3	The label should be placed right of centre allowing a minimum of **5 cm** from either edge.
4	The label should not be placed over a seam nor should sealing bands be placed over the label in a manner that interferes with the scanning of the label.
5	The bottom edge of the label should be within the range of **40 cm to 80 cm** from the bottom of the pallet.
6	If the pallet is less than 50 cm in height, the label should be placed as high as possible on the pallet considering the minimum distance of 5 cm from either edge.

See figure below.

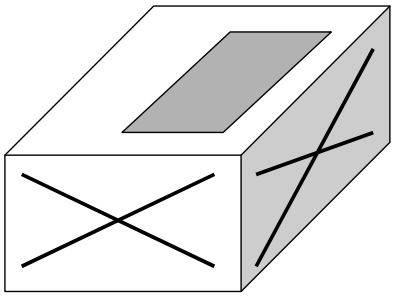


10.1.2 Transport packages

The following requirements should be followed :

1	Pieces **up to 1 m** in height require the placement of the label on top of the package.
2	Transport packages **greater than 1 m** in height should follow the recommendation of
3	The placement on the side of the package should be avoided whenever possible; it must not be placed on the bottom at all.
4	The edges of the label should be at least **2,5 cm** away from the edges of the package

.



.

11 Appendix : Local Temporary Exceptions

Below list of exceptions are related to the routing barcode and have been agreed globally in order to ensure the current specifications also reflect the latest modifications adopted in the divisional red DHL NSS 21.3.2.0.

These exceptions will be absorbed when the target specifications 2.6 are released.

11.1 Routing Barcode

11.1.1 General

The Routing Barcode is not mandatory in AM.

(Only the target solution in 2.6 will require automatic sorting based on a routing barcode)

11.1.2 Feature Section

The feature section in the routing information is not mandatory in AM/AP/EMA. Where the product features cannot be printed “000000” will be put in the feature field.

The customs flag is the only mandatory feature which should be available across all printing applications. As a consequence, the last three digits of the feature section will be printed as “001” for dutiable shipments and “000” for non-dutiable shipments.

11.1.3 Destination Postcode

The following rule only applies to red DHL products and applications.

The blank/missing postcode part in the barcode must be replaced by a DHL valid 3-letter destination code, preceded by a colon “:”

Examples :

- (2L)LB:BEY+42101073 (LB stands for the Lebanon and :BEY is the destination service area code for Beyrout)
- for Ireland “:ORK” will be used as destination service area code for the city Cork

12 SECTION B : BACKGROUND INFORMATION

13 Archival Label Copy

13.1 Requirements

DHL is required to keep archive copies of all Air waybills (AWB's) in order to meet legal and customer service access requirements. **This however only applies to air-based shipments.**

Air waybills are normally scanned into an **imaging system (PCDOC).** In addition, Air waybills for dutiable shipments are scanned into the **CIA system** to support customs clearance processes.

When an AWB does not exist (e.g. electronic via EDI, EasyShip, etc), there is no physical Air waybill. In this instance a copy of the Network label is normally kept. The current e-com applications print out an extra copy of the network label so that one can be stuck on the shipment and one can be saved for scanning.

To improve this process some countries make use of a technique called **COLD LOADING**, where a feed of the E-COM data is loaded into the archival system. This data can subsequently be used to generate a reproduction of the label image.

The COLD technique eliminates the need for scanning of the labels for non-dutiable shipments, but because of the requirement to scan Air waybills and related customs paperwork into CIA, a hardcopy label with the Air waybill barcode is still required. Without this Air waybill hardcopy, the customs paperwork would have to be manually sorted and indexed.

In the new scenario where a unique label will be generated for each piece of a multi-piece shipment, there is still a requirement for archival of the Air waybill level information. Only one label / Air waybill has to be archived for each shipment regardless of the number of pieces in the shipment. The piece id that is barcoded onto the network label is not required for the archive.

Countries that use COLD LOADING importing might not require archive copies for non-dutiable shipments, but will require them for dutiable shipments to support CIA scanning.

Archive Label Example

ARCHIVE COPY
NOT TO BE ATTACHED TO SHIPMENT

FROM : Abbott & Abbott S.A./N.V.
Mr Smith
Maintenance Dept, Building Seven
Brussels National Airport
1930 Zaventem, Belgium

ORIGIN:
BRU

TO: MAYERS & GREENE CORPORATION
Ms Marianna WOJCIECHOWSKA
WORLD CENTER BLDG.
FLOOR. 6
330 SEN GIL PUYAT AVENUE
2100 MAKATI CITY, MANILA
PHILIPPINES Tel: 63-2-895-0511

DG-SPD-ND-SB-CD

Time
B

Date : **26/12/2003**
Destination : **MNL**
Pieces : **2**

Weight : **1,500 Kg**
Product : **WPX**
Service : **DDU**

Description. : **High Pressure Hydraulic Machine Parts**
Shipment Number : **8741493123**
Sender's Ref : **Braun25017**
Sender's VAT : **155 9856 02**
Senders Phone: **+32 2 203 617 428**
Bill to Account: **55443 48567 44**

Customs Value : **2,000,000 EUR**
Imp/Exp Type : **Permanent**
Insurance Value: **3,500,000 EUR**
Terms of Trade:
Reason for Export

AIRWAYBILL: **8741493121**

L1000

14 Standard barcode specifications

14.1 Code 128

Code 128 was introduced in 1981 as an alphanumeric symbology with a very high density.

The characters are encoded by a continuous code with a variable length that comprises stripes and spaces of differing widths.

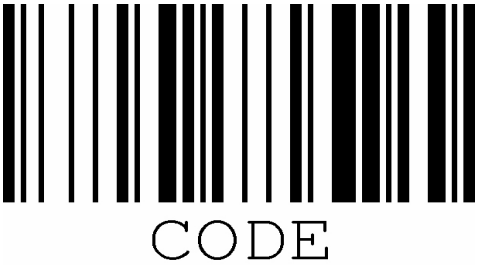
Every code 128 character has 11 modules, each of which can be black or white. Every character has 3 stripes and 3 spaces; code 128 is therefore a (11,3) code.

Example of a code 128 character :

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |10 |11 |

The character “CODE” will be barcoded as follows

START	C	O	D	E	STOP



Code 128 encodes the full 128-character ASCII character set using three alternate character sets, A, B, C. So, a given character pattern can have several meanings, depending on the character set used.

Every code 128 barcode has a control character which is the modulus 103 sum of the value of each character is multiplied by a weighting factor.

The bar codes for the License Plate and Routing Bar code must meet the following specifications.

All barcode parameters must be as defined by the latest version of the “Uniform Symbology Specification - USS CODE 128” published by A.I.M. International.

1. **X-Dimension**	X	Between 0.370 mm to 0.425 mm
2. **Quiet Zones**	Q	6mm at each end of symbol
3. **Symbol Height**		25mm
4. **Code Orientation**		Picket fence (standing).
5. **Encoding**		Character sets must be used to minimise the physical length of the bar code. Character set **C** must be used to encode groups of numeric data when this will reduce the length of the bar code.
6. **Check Character**		**As defined in Uniform Symbology Specification - USS CODE 128, see above**
7. **Symbol Quality**		CEN Grade **3** or better when measured according to EN 1635

14.2 EAN 128

EAN 128 is a variation of code 128. EAN 128 uses an additional function character.

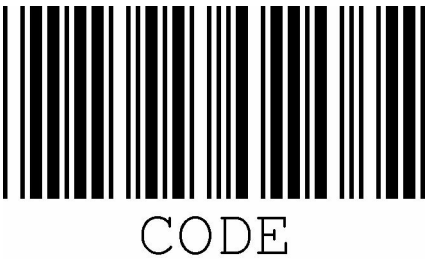
If a FNC1 character is used directly after the start symbol then this indicates that the following data must be translated as defined by the Uniform Code Council (UCC) and EAN International. It is used for all EAN application identifiers, such as the SSCC or the routing information.

14.3 Code 39

Code 39 was the first alphanumeric symbology developed. Every code 39 character has five bars and four spaces, making a total of 9 elements. Of these 9 elements, three are wide and six are narrow. This symbology is sometimes referred to as “3 of 9 Code”

The character “CODE” will be barcoded as follows :

START	C	O	D	E	STOP



The bar codes for the AWB number must meet the following specifications.

All **BAR CODE PARAMETERS** must be as defined by the latest version of the “International Symbology Specification - ISS **CODE 39**” published by A.I.M. International.

1. **X-Dimension**	x	Between 0.370 mm and 0.425 mm
2. **Wide To Narrow Ratio**	N	Between 2.2 and 3.0
3. **Inter-character Gap**	I	Between X-t and X+t

4. **Quiet Zones**	Q	mm, at each end of symbol
5. **Code Orientation**		Picket fence (standing)
6. **SYMBOL QUALITY**:		CEN Grade **3** or better when measured according to EN 1635
7. **SYMBOL DIMENSIONS:**		
AWB Number Bar Code:		Height: 25 mm ± 0.5 mm
(10 numeric characters)		Width: 72 to 93 mm *(inclusive of quiet zones and dependant on bar code characteristics)*
Bar Code Length (L):		(C+2) (3N+6) x + (c+1) I + 2Q *where c = number of characters*

15 Linear & Two dimensional barcodes

15.1 Human readable data form

To ensure an error free Ideally, the label could only consist of only the piece identifier in bar-coded form. This would allow the identification of the package by scanning of this code. All other relevant data (ship to address etc.) could then be retrieved from databases.

handling, to reduce the dependency on readability of barcodes and information stored in computer files and to simplify operations, additional information in bar-coded- and human readable form is added to the label as well.

Some information is only presented in human readable form. All linear bar-coded information is represented in human readable form as well.

15.2 Linear barcode

Printed linear barcodes are widely used for the encoding and capture of data: they are easy and economic to apply and can be read, with a high degree of accuracy, using relatively low cost readers.

A barcode is a symbol that comprises a series of parallel stripes and open spaces of differing width. Defined patterns of lines and spaces are used to represent the symbol (a symbol is a combination of a number of narrow and wider black and white stripes).

To read the information that is in a barcode a scanning apparatus, like a reading pen, is moved over the symbol. This scan-apparatus analyses the content of the symbol by ‘reading’ the line pattern and the original data can be displayed. The technology of the read apparatus is constantly being developed for use in all sorts of situations and circumstances.

The barcode is an automatic identification technology. By that it is possible to collect existing data quickly and without error. However, barcodes themselves do not solve any problems. The combination of barcodes with good computer hardware and software applications creates the possibility to improve the quality of the work, the productivity and eventually the profitability.

The label contains two linear barcodes. The data in the linear barcodes is preceded by ANS Data Identifiers (DI) or EAN Application Identifiers (AI), to identify the data in the barcodes. In this way the scanning software can distinguish between the different barcodes on the label. This technique also allows putting more than one piece of information into one linear barcode, each one headed by a DI/AI, with a separator char in between. For ANSI codes, the

standard separator is a blank (“ “). For this reason, in ANSI barcodes the data itself may never contain a blank.

15.3 Two dimensional barcode (PDF417)

15.3.1 General information

The two-dimensional barcode, represented by the PDF417 barcode symbology, can be used to print the complete transport order in one barcode on the label.

The shipment data that belong to the associated shipping unit are recorded in a two-dimensional barcode. These shipping data are placed in the barcode according to a certain layout. A two dimensional barcode is constructed from diverse rows while the one-dimensional barcode is constructed from a row of stripes.

! The availability of this feature may vary from region to region. Therefore a bilateral agreement between customer and his DHL contact is necessary.

There are different barcode symbologies (‘languages’) for the production of the two dimensional barcode.

15.3.2 PDF417 types

A PDF417 is a two-dimensional barcode, accepted by both ANSI and CEN as ‘open system technology’ standard.

Two dimensional barcode symbologies are increasingly used for applications in which the volume of data to be encoded exceeds that which can be conveniently represented in linear barcode form. They provide increased data density at the cost of requiring a greater degree of control over the printing process and the use of more sophisticated, and thus more costly, reading equipment.

The content of the message is an EDI IFTMIN message.

There are three sorts of PDF417 codes :

the Micro PDF,

the Macro PDF

the Standard PDF

The difference is in the quantity of data that can be stored.

The Micro PDF is not suitable because the storage capacity is too small. On the other hand, the Macro PDF is so large that processing the data costs too much time. In a standard PDF417 code up to a maximum of 1800 characters can be stored, depending on the error correction level.

ISO has defined PDF as a standard.

15.3.3 Message structure within PDF417 barcode

The transport order data is structured in an Order/Instruction message according to the current version of the Deutsche Post World Net Message Implementation Guide for IFTMIN (Label) Message.

15.3.4 Message encoding

Encoding shall be done according to ***ISO 15434*** Transfer Syntax standard, using format indicator 02 (indicating the format "Complete EDI message / transaction"). It means that the whole message is encoded rather than replacing the UN/EDIFACT segmenting by Transfer Syntax format headers. As a result, the whole message can easily be encoded in PDF417 following this standard and can subsequently be directly passed on to an EDIFACT translator after scanning.

15.3.5 EDIFACT message coding

The EDIFACT message is to be coded into a PDF 417 barcode according to PDF 417 standard ***ISO 15438.***

15.3.6 Dimensions

Naturally the quantity of information determines the size of the barcode. A number of constraints need to be considered:

Quiet Zone	minimal 2 times the X-dimension
Row and Column Height	An individual row is 0.75 mm high. The number of rows and columns can vary. To provide a fast scan processing 30 to 40 rows are necessary. The number of columns has a direct relationship with the number of rows. After all, the more rows that are used, the fewer columns are necessary to contain the same information. In practice the number of rows can vary between 30 and 40, while the number of columns varies between 7 and 18, excluding row indications

X Dimension	Each bar's width must be a multiple of the smallest bar width. For instance, if the narrowest bar is 10 mil, then possible bar widths can be 10 mil, 20 mil, 30 mil, 40 mil, etc... But not, 15mil, or 25mil since these sizes are not multiples of the narrowest bar size of 10 mil.
Dimensions	a maximum of 850 bytes allowed max. 100 mm wide and 35 mm high, of course depending on the size of the label. The dimensions of a PDF barcode depend on the content. The more bytes, the larger the barcode. When working with fixed dimensions a barcode with a lot of 'air' is formed. This means a worse scan quality and effectively a lower security level, as the calculation of code-words includes the 'air'.

15.3.7 Structure

Like other barcode symbologies, PDF417 has a well defined physical structure.

Below table describes each of these structural components:

Start/Stop pattern	Every barcode has a start pattern on the left, and a stop pattern on the right. These patterns are unique for each type of barcode
Code Words	In a PDF417 barcode, each bar and space does not store data. Data is actually stored in code words. A code word is a consecutive sequence of 4 bars and 4 spaces totalling 17X wide. The 417 in PDF417 refers to this code word structure Code words reside between the start pattern on the left, and the stop pattern on the right. This region contains several types of code words including data code words, control code words, and row indicator code words.
Row Indicator Code words	Adjacent to the start and stop patterns, are PDF's right and left row indicators. These indicators are code words that are used to store information required to decode the PDF417 symbol. This includes the row number, number or rows, number of columns, and the error correction level used.
Data Code Word Region	User data is first encoded into code word values. These code word values are then converted into physical code words represented by bars and spaces as described above. Data code words are physically located between the left and right row indicator code words. To the right, you can see the data code word region:
Rows	PDF417 symbol is made up of multiple rows. A PDF417 barcode can have anywhere from 3 to 90 rows. This allows a PDF417 symbol to be reshaped by adjusting the number of rows.
Columns	A PDF417 symbol is made up of multiple data columns, which are sometimes referred to as the data column area. The number of data columns can vary from 3 to 30, to accommodate user's real estate requirements. These columns contain encoded data, as well as error correction information. Within the data column area, a single PDF417 can contain no more than 928 code words.

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15.4 Linear & Two dimensional checksum methods

The printing and reading of barcodes and 2D symbols is a form of data communications in which one party, the printer, “sends” data to one or more others, who “receive” the transmission by reading the printed code. As with any data communications application, physical disturbances and noise may intervene, resulting in a difference between what was (intended to be) printed or sent and what is actually captured by the reader or receiver.

In the environment considered, such disturbances may include imperfections in the printing, physical damage to the supporting material or printed mark, interference between the ink and the substrate to which it is applied and physical limitations of the reader. Such errors are generally referred to as data capture or read errors, even though they may actually result from imperfections in the printing.

To protect against the occurrence of data capture errors, the data which are encoded and read need to include a sufficient degree of redundancy to allow the reader to detect, and if possible, correct any data capture errors which occur. This redundancy is usually coded in the form of a check-digit.

Redundancy inherent in the encoded data itself.

Two following two standards bodies have proposed their own check digit methods :

- **EAN** : The 18th digit of the LP is always the check digit. It is part of the License Plate.
- **UPU** : A check digit calculation is available but is not mandatory. It is not part of the License Plate, to be printed in plain text only, not in the barcode. There is also a control number calculated by the application. This is calculated following the modulus 10 method. The check digit number is not part of the License Plate.

Basically, there are three levels of error-detection :

15.4.1 Redundancy inherent in the encoded data itself

An example is the check-digit at the end of the EAN SSCC License Plate. The check-digit has to be calculated by the user (the sender of the data) and incorporated into the data. This check-digit is part of the data and will be stored with the data. This way of error-detection can be used both with bar-coded- as well as human readable representations of the data.

This is an example of check digit calculation which uses a weighted Modulo 10 scheme for the String 1234567:

Characters	1	2	3	4	5	6	7
Value	1	2	3	4	5	6	7
Weight factor	3	1	3	1	3	1	3
Product	3	2	9	4	15	6	21

The sum of the product of the characters values and their weight factors is:

3+2+9+4+15+6+21 = 60

remainder of 60/10 = 0

check digit = 10 - 0 = 0

bar-coded string = **12345670**

15.4.2 Redundancy inherent in the barcode symbology

Examples are the check-digit in the Code 128 linear barcode symbology or the error-correction possibilities of the PDF417 two-dimensional barcode symbology.

This way of error-detection is not part of the encoded data itself. It is usually automatically generated at the time of creation of the barcode (without the sender's interference), and is automatically decoded and verified by the scanning hardware when capturing the data. As a result of this, the capture of data from linear as well as two-dimensional barcodes is highly reliable. This way of error-detection can only be used with bar-coded representations of the data.

15.4.2.1 Code 128 check digit

The control character (check digit) follows the data and directly before stop character. The control character is the character that is equal to the modulus 103 sum of the value of the start character and the weight factor of the characters that follow the start character. The weight factor increases in order 1,2,3,4,.., starting with the first character that comes after the start character. As an example, the control character of a symbol with content "VGL128" is calculated as follows:

Characters	Start B	V	G	L	1	2	8
Value	104	54	39	44	17	18	24
Weight factor	1	1	2	3	4	5	6
Product	104	54	78	132	68	90	144

The sum of the product of the character values and the weight factor is:

104+54+78+132+68+90+144 = 670

Dividing 670 by 103 gives 6, with remainder 52. The control character is T, which has the value 52.

15.4.2.2 PDF417 Correction Capability

The PDF417 symbology has error correction capability. This capability enables scanners to read the barcode even if it has been torn, written on, or damaged in other ways. How much damage a symbol can withstand depends on the amount of error correction in each PDF417 symbol. The user has the ability to select 1 of 9 error correction levels for each symbol printed.

Error correction is specified by selecting a level from 0 to 8. At level 0, a damaged PDF417 cannot be read, but the damage can be detected. At levels 1 through 8, a PDF417 symbol can still be read, even when damaged.

As the error correction level increases, more damage can occur to the symbol and still be read. Consequently, the higher the error correction level, the larger the symbol becomes, while the data capacity goes down. The following table illustrates these facts:

PDF417 symbols can be damaged and still decoded. The amount of damage that a symbol can withstand is its error correction capacity. For example, at level 5 error correction, 64 code words of error correction are used. At this level, 31 code words can have errors, while still being read correctly. If more than 31 errors exist, the symbol is unreadable.

DHL uses level 5.

15.4.3 Redundancy added to the human readable interpretation of the data

An example is the error detection code (check-digit) presented in the UPU standard S26 “License Plates for parcels”. This way of error-detection is not part of the encoded data itself. The check-digit has to be calculated by the user (the

sender of the data), and added to the human readable representation only (the bar-coded representation does not change). It can only be used with the human readable representation of the data.

16 List of IAC codes

The updated list of IAC codes are available in the L&I Master Data file or directly from the official NNI site (**www.nen.nl**)

IAC	Issuing agency
0 thru 9	EAN International- European Article Numbering Association
D	NATO AC/135
J	Universal Postal Union
KNO	Norstella
LA	JIPDEC/CII – Japan Information processing Development Corporation / Electronic Commerce Promotion Center
LB	Telchordia Technologies, Inc.
LE	EDIFICE – Electronic Data Interchange for Companies with Interest in Computing and Electronics
LF	FIATA – International Federation of Freight Forwarders Associations
LH	EHIBCC – European Health Industry Business Communications Council
ND	DHL Express Germany (former Danzas DE)
NL	TPG POST
OD	ODETTE EUROPE
PA	CEPI – Confederation of European Paper Industries
SI	SIEMENS
ST	EUROFER – European Confederation of Iron and Steel Industries
UN	Dun & Bradstreet
UT	EuroExpress – Verein EuroExpress
VGL	DHL Express Benelux (former van Gend & Loos)
VIB	IBM EMEA – International Business Machines

17 Normative references

17.1 ISO standards

In this document, references are made to the following international standards defined by the ISO organization (International Organization for Standardization), as far as they are relevant for the standardised DHL label and identifier.

Summary Table of ISO standards used

ISO #	CONTENT
ISO 3166	unique identification coding of countries
ISO 15394	Packaging – Barcode and two-dimensional symbols for shipping, transport and receiving labels Describes minimum requirements for the design of labels containing linear barcode and two-dimensional symbols on transport units
ISO 15416	Information Technology – Automatic identification and data capture techniques – Barcode print quality test specification – Linear symbols
ISO 15417	Barcode symbology specification of the linear barcode Code **128**
ISO 15418	Information Technology : **EAN/UCC Application Identifiers** and **ANS MH 10 Data Identifiers** and Maintenance
ISO 15420	Barcode symbology specifications for **EAN** / **U.P.C. (EAN 128)**
ISO 15434	Information technology – Transfer syntax for high capacity ADC media
ISO 15438	Barcode symbology specifications for **PDF417**
ISO 15459-1	Specifications for unique identification of transport units – Part 1 General
ISO 15459-2	Specifications for unique identification of transport units – Part 2 Registration procedures
ISO 16388	Barcode symbology specifications for **Code 39**.

17.2 CEN standards

In this document, references are made to the following international standards defined by the CEN organization (*Comité Européen de Normalisation*):

#	CONTENT
EN 1573	**Multi Industry Transport Label** Definition and usage of the Multi Industry Transport label, including License Plate usage.
EN 1635	**Barcoding -Test Specifications for Bar Code Symbols** Specifies quality levels for printed linear barcodes.

17.3 UPU standards

In this document, references are made to the following international standards defined by the UPU organisation (*Universal Postal Union*):

"UPU S26-4 – License Plates for parcels" defines the UPU implementation of ISO/IEC 15459 and an associated label for the identification of parcels

17.4 Industry standards

17.4.1 MH10.8.2

This standard has been defined by the ANSI organisation (*American National Standards Institute)* as the "Mother of modern label layout standards". International labelling standards such as CEN 1573 or ISO 15394 are based on the concepts of this standard.

17.4.2 EAN.UCC

This standard describes the system of EAN Application Identifiers (to complement ISO 15418), describes usage and structure of EAN SSCC in digital form and in barcoded form, describes further UCC/EAN standards, such as the Global Location Numbers (GLNs), describes format and usage guidelines for the UCC/EAN Logistics Label.

18 Terms and definitions

The following terms and definitions are used within this document:

Please also refer to the DHL Glossary for more definitions of business terms :

Term	Definition
2D Barcode	Machine readable code, capable of storing large amounts of data, which must be examined both vertically and horizontally to read the entire message. Two dimensional symbols are one of two types of machine readable symbols: matrix symbols and multi-row symbols. 2D symbols have error detection and may include error correction.
A5 eCom label	A version of the shipment label produced by the customer on part of a standard paper sheet, using a laser printer
ADC	Automatic Data Capture
ANSI	American National Standards Institute
Application Identifier	A UCC/EAN specified character (or string of characters) that defines the general category or intended use of the data that follows
Automated Label	A label generated by a computer system. It is either produced by DHL, or by the customer according to DHL's specifications.
Baby Label	A smaller version of an automated label containing at least the License Plate. Baby Labels are produced to label several pieces of the same shipment to avoid unnecessary replication of the whole label content. Each baby label has a different License Plate.
Barcode	A machine-readable design in the form of a pattern of vertical stripes identifying a piece
CEN	Comité Européen de Normalisation ; European Standards Institute
CEN Grade	An overall, weighted measure of the quality of the tested bar code. This grading is calculated from a number of different parameters. Grade 3 is the cut-off point of acceptable readability; 4 is the highest grade. Bar codes with a CEN grade below 3 is not acceptabl
Check digit	Digit added to a barcode for verification purposes. The check digit is usually calculated from the barcode content using a modulo algorithm. A check digit can be added to the content of a barcode, and/or it can be part of the barcode symbology (in the latter case ist value is not

Term	Definition
	transferred by the scanning device).
CMR	The CMR logo (Convention des Marchandises Routiers) is used for road freight between many European countries. This section is mandatory for all international road-based shipments.
Code 128	Code 128 is an alphanumeric barcode symbology with high density. It is a continuous code with a variable length that comprises elements of differing widths. It contains a check digit.
Code 39	Alphanumeric barcode symbology with relatively low information-density. It contains a check digit.
Common Label	See DHL Transport Label
Data Identifier	A specified character (or string of characters) that defines the general category or intended use of the data that follows
DHL Transport Label	The standard DHL label used for transportation and issued in accordance with global DHL label specifications and ISO 15394 and 15459. It contains a piece identifier (the licence plate), operational and customer information.
EAN	EAN International, former European Article Numbering organisation, is the Issuing Agency for EAN License Plates, the so-called SSCC.
EAN128	EAN 128 is a sub-standard of Code 128, defined by EAN International and specified in EN 799.
Fence orientation	When a barcode is printed in portrait orientation, this is called fence orientation, as the bars are standing like a garden fence. For thermal printer this is always the preferred method.
IAC	Issuing Agency Code
Inter-character gap	The space between the last bar of one character and the first bar of the next character in a bar code.
ISO	International Organisation for Standardisation
Label	An adhesive piece of paper attached to a piece to indicate the nature, ownership, destination, contents and/or other particulars of the piece.
Ladder	When a barcode is printed in landscape orientation, this is called ladder

Term	Definition
orientation	orientation, as the bars form a kind of a ladder. For thermal printer this method is never recommended.
License Plate	A code assigned to a transport unit by ist issuer, in accordance with ISO/IEC 15459-1:1999, *Technical Standard for unique identification of transport unit*s
Linear barcode	One dimensional barcode.
Manifest	Pick-up list provided by customer. It contains all necessary order data for DHL, especially the License Plate of each piece to be shipped.
Manual Label	See Manual Label
Manual Waybill	A pre-printed multi-ply transport order form that is used as a DHL transport label. It is filled out by the shipper or a third party with information about the piece and the desired service.
Network Label	An adhesive-backed form that contains an extract of shipment details in a bar code and a human-readable format
OCR	Optical Character Recognition; Machine interpretation of human readable information
PDF417	A PDF417 is a two-dimensional barcode, accepted by both ANSI and CEN as 'open system technology' standard.
Print tolerance	The maximum allowed width deviation from the normal value. If a narrow element is wider than X + t, it will be interpreted as a wide one.
Quiet Zone	A clear white space of a minimum 6 mm wide, containing no marks, at each end of the bar code.
Routing	A sequencial list of connections between facilities, describing a transport between two locations.
Routing Label	A small label separately produced for DHL internal use which has to be attached to the original label by DHL whenever the existing DHL Transport Label does not contain a proper routing code.
Shipment Label	An adhesive backed label that establishes a contract between a customer and DHL for the movement of a shipment.
SSCC	Serial Shipping Container Code, License Plate issued by EAN International.

Term	Definition
Symbology	A set of rules defining the way information is presented in a barcode. Examples are Code 128, EAN128 and Code 39
UCC	“Uniform Code Council”; the American counterpart of EAN International.
UPU	“Universal Postal Union” Department of the United Nations, co-ordinating world-wide postal services.
Waybill	A waybill contains order data, billing data and the transportation data (routing data and special handling instructions) on one label. A waybill accompanies the shipment from sender to receiver.
Wide to Narrow Ratio	Wider elements in bar codes are multiples (ratios) of the minimum width. The x-dimension is the cornerstone for measurement of the wide-to-narrow element ratio and to symbol readability itself as it relates to varying element widths.
x-Dimension	The width of the smallest bar in a one-dimensional barcode

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